

WEATHER & RETAIL

CAN WEATHER HELP PREDICT RETAIL PURCHASING BEHAVIOUR?

PROCESS



981,000+
UK Retail
Transactions
(2009-2011)

+



UK Daily Weather
Data
(2009-2011)

→



Data Mining
(Clustering,
Association
Rules)

→



Actionable
Retail Insights

Average unit price purchased during
warm and dry conditions



33%

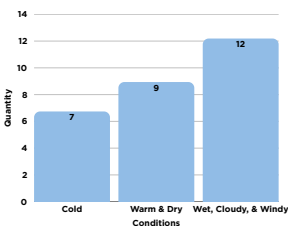
higher than cold
conditions

HIGHER-PRICED PURCHASES



Promote higher-priced products
during warm, dry weather

HIGHER QUANTITY PURCHASES



WET, CLOUDY
& WINDY
WEATHER



LARGER
PURCHASES

Prepare inventory when wet, cloudy &
windy weather is forecast

**WEATHER HELPS — BUT
CAN'T TELL THE WHOLE
STORY**



~35%

**PREDICTION
ACCURACY**

Weather and timing alone
were not enough to reliably
predict transaction sales.

Average quantity purchased during
wet, cloudy & windy conditions



81%

higher than cold
conditions

Under Low Temperature + Low
Rainfall Conditions



28%

stronger purchasing
association with White
Hanging Heart
T-Light Holder product

What should Retailers Do?

1

Plan Inventory

Prepare for larger
purchases
during wet weather

2

Adjust Promotions

Promote higher-
priced products
during warm, dry
weather

3

Cross-Promote Products

Recommend
frequently
purchased-together
products

4

Use Multiple Factors

Combine weather
with other
factors to predict
sales